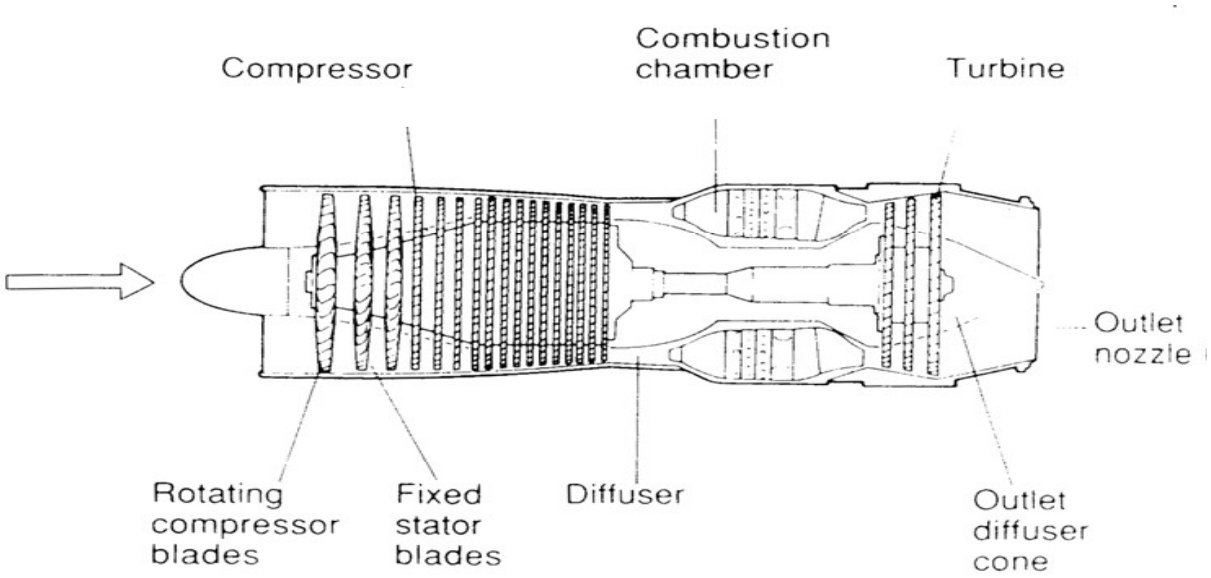
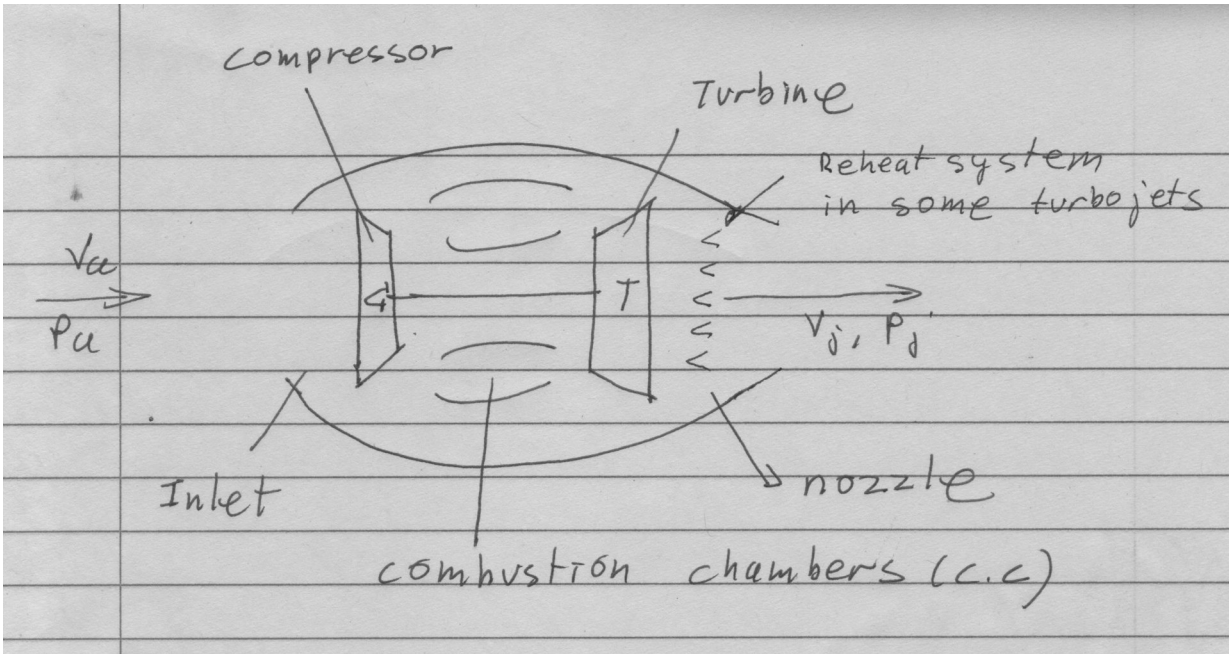
2. Jet Engines

2.1 Classification

Turbojet:



Schematic sketch



Turbojet components:

Intake (inlet):

It slows the flow with minimum loss in stagnation pressure, stagnation temperature can be treated as preserved.

Compressor:

It raises stagnation pressure, hopefully in a process that is close to isentropic process. The goals:

1. To generate a favourable pressure gradient over the nozzle
2. To yield good combustion efficiency
3. To help having sufficient air mass flow rate

Combustion chamber:

It raises stagnation temperature (~2000K) by burning fuel with minimum stagnation pressure loss.

Turbine:

It expands the flow by converting some of the thermal energy to mechanical energy in order to power the compressor through the connecting shaft. Best design aims towards a process close to an isentropic process.

Reheat system:

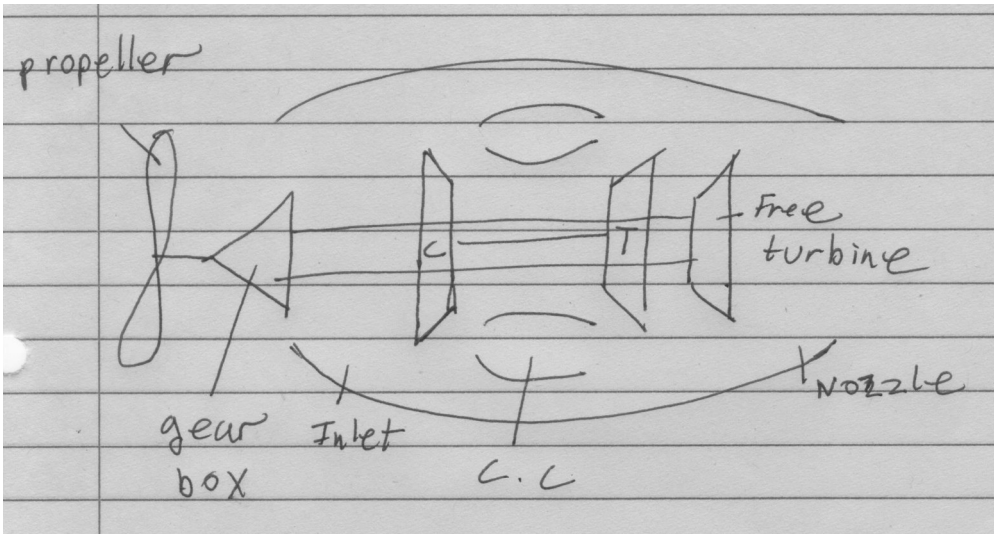
It exists in some engines as the Olympus that powered the Concorde. Because there are no moving parts (unlike the turbine), we can increase significantly the stagnation temperature (to ~5000K). However, it has low efficiency and there is a risk of thermal choking.

Nozzle:

It converts thermal energy to kinetic energy,

1. Convergent nozzle: It can accelerate the gas up to $M_j=1$. If $M_j<1$, $p_j=p_a$. High stagnation to ambient pressure ratio p_0/p_a will lead to a choked nozzle and $p_j>p_a$.
2. Convergent-divergent nozzle: It can achieve $M_j>1$ and even a perfectly expanded jet of $p_j=p_a$ when a variable nozzle geometry is used.

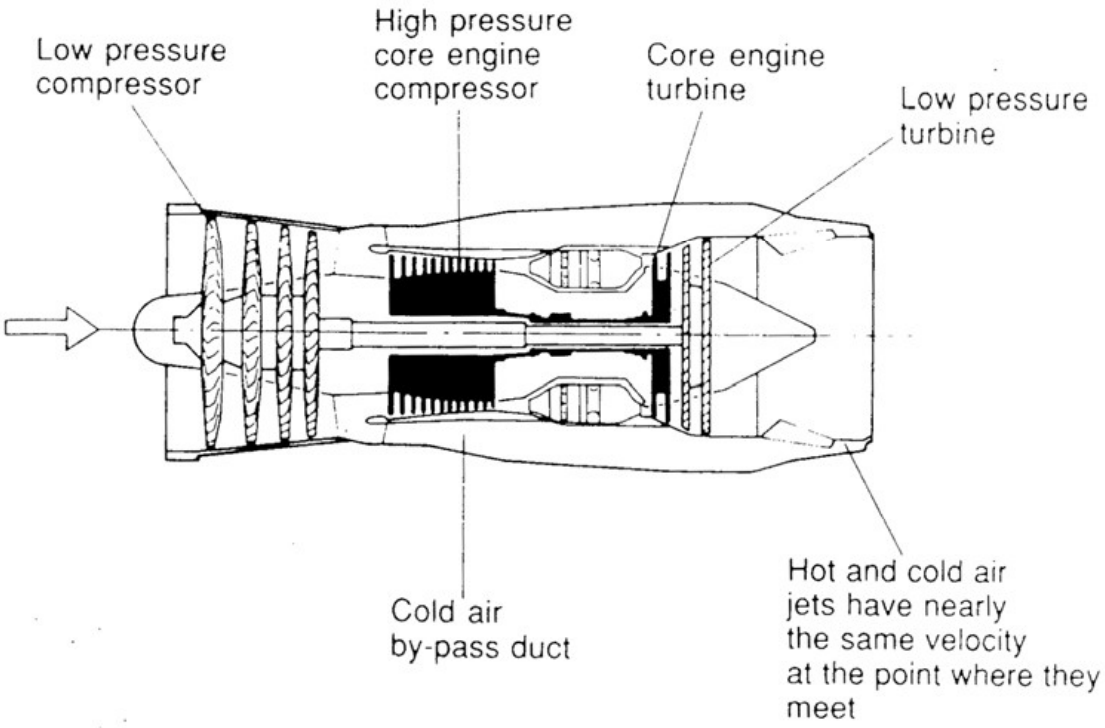
Turboprop:



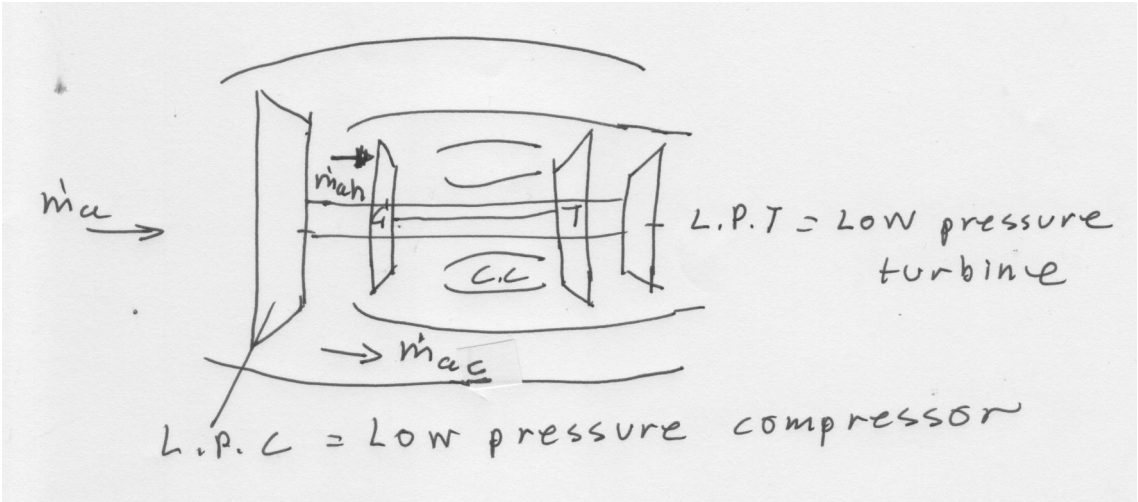
The core is the turbojet. The free turbine after the high pressure turbine, provides energy to the gear box (which reduces the RPM) that powers the propeller.

Turbo-shaft:
Similar operation as turboprop, it is used for helicopters and heavy land vehicles

Low by pass engine:



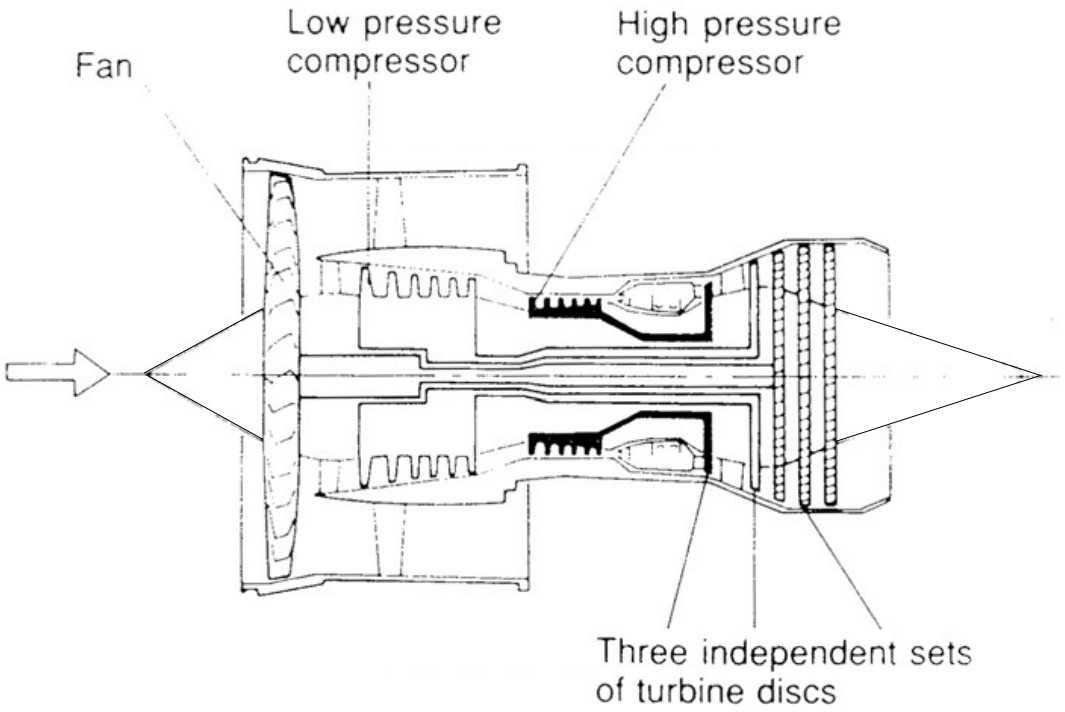
Idea: increase air mass flow rate and thus propulsive efficiency



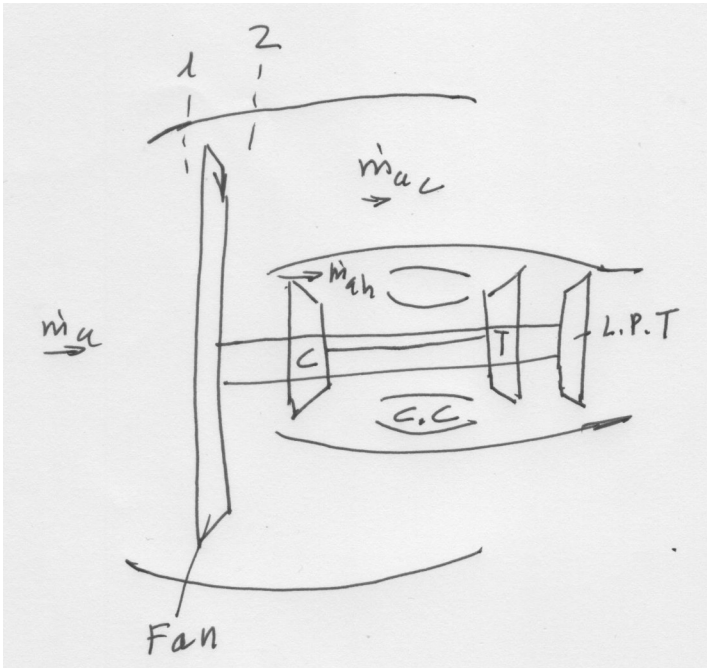
$$\dot{m}_a = \dot{m}_{ah} + \dot{m}_{ac}$$

$$\text{By pass ratio} = \text{BPR} = \mu = \frac{\dot{m}_{ac}}{\dot{m}_{ah}}$$

High by pass ratio:



Schematic description

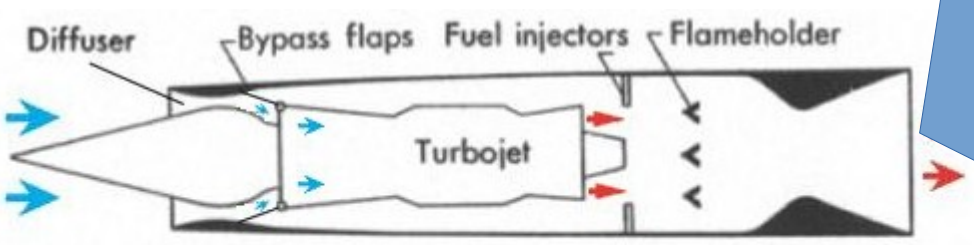


BPR > 10, the choice for subsonic airliners

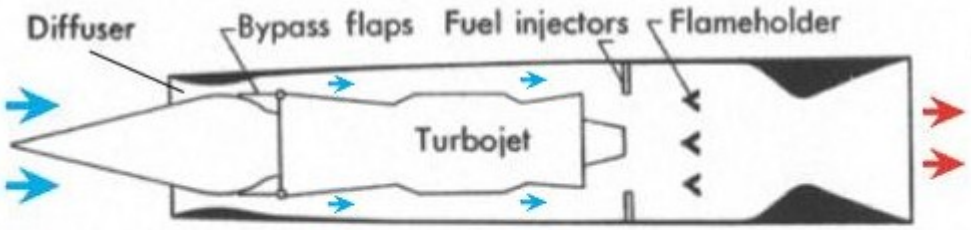
$1< p_{02}/p_{01} < 2$, high propulsive efficiency and low jet noise, but the engine is not suitable for supersonic flight due to high wave drag.

Ramjet:



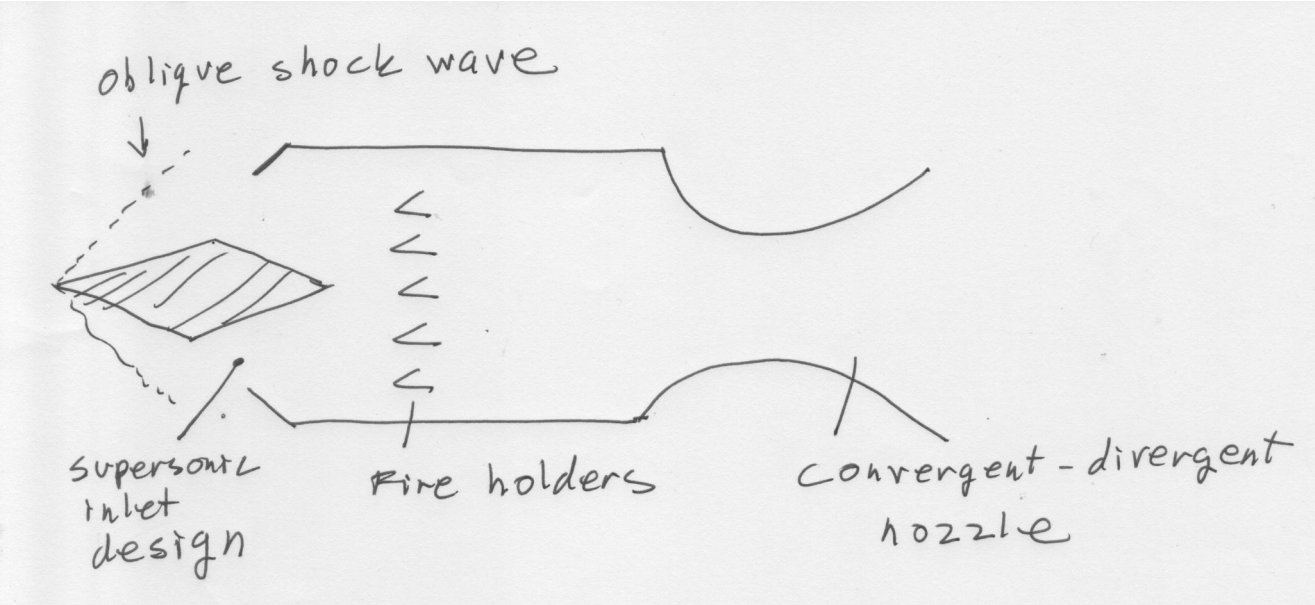


a) Bypass flaps allow flow into turbojet



b) Bypass flaps block flow into turbojet during ramjet mode

Schematic sketch - ramjet

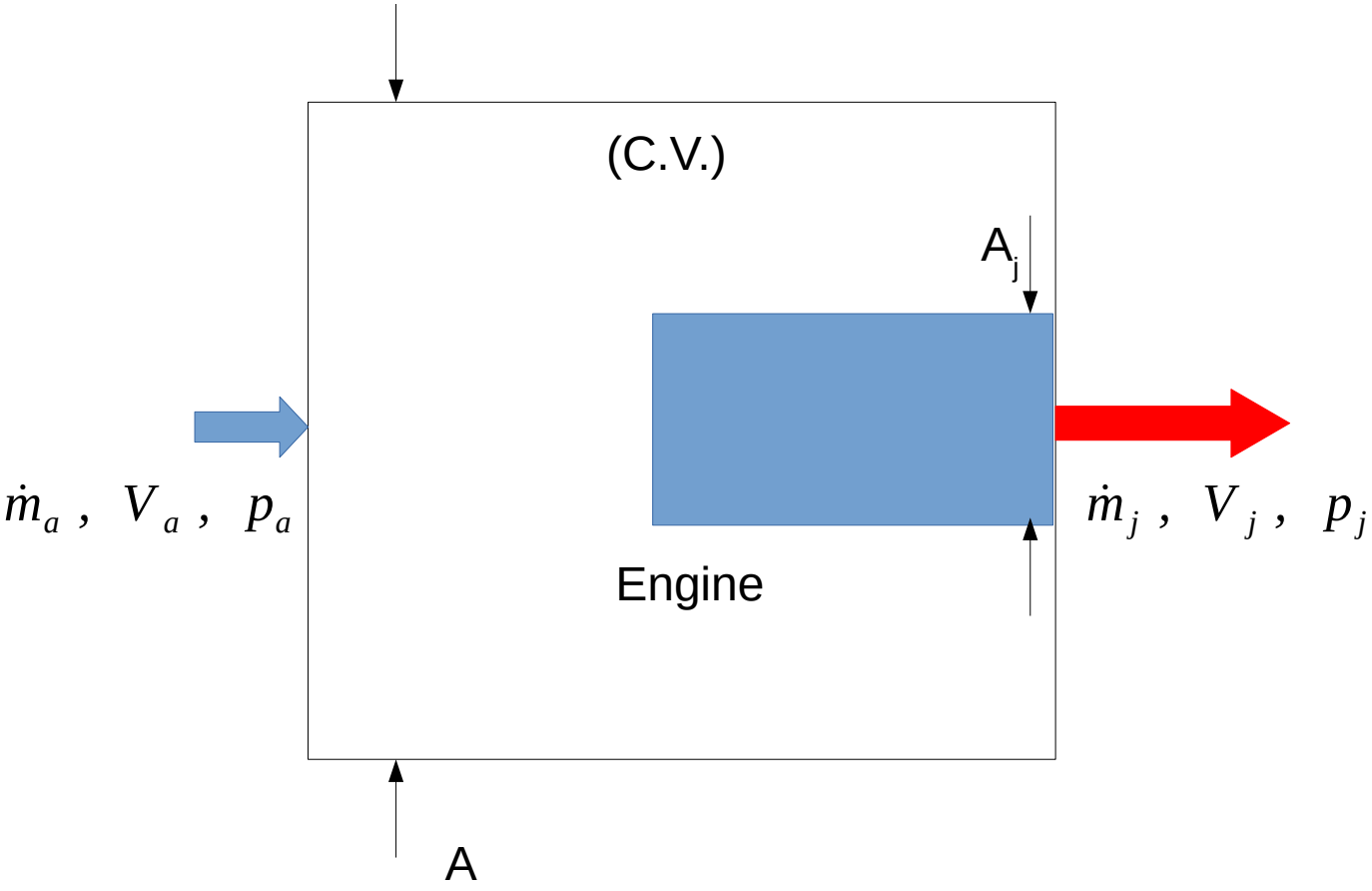


No turbine, so combustion temperature can be raised well above 2000K to about 5000K. However, because there is no turbine there is no compressor. The air is compressed by the ram effect, i.e. bringing the flow to a halt in a process that is close as much as possible to an isentropic process.

So the ramjet engine can work only during flight and is effective in supersonic flight (M_a>1). Hence we must make sure that there are no normal shock waves in the inlet. One way is to use a cone as illustrated above and is seen in the SR71 aircraft. The oblique shock wave must not enter the inlet and thus the best practice is that the oblique shock wave almost touches the inlet lip but does not enter it.

2.2 Thrust and propulsive efficiency

2.2.1 The turbojet



Momentum equation:

$$\dot{m}_a V_a + p_a A + F_n = \dot{m}_j V_j + p_a \left(A - A_j\right) + p_j A_j$$

$$F_n = \dot{m}_j V_j - \dot{m}_a V_a + \left(p_j - p_a\right) A_j$$

F_n is the net thrust

$$F_g = \dot{m}_j V_j + \left(p_j - p_a\right) A_j$$

F_g is the gross thrust and where $\dot{m}_j = \dot{m}_a + \dot{m}_f$

$\dot{m}_f$ is the fuel mass flow rate

Define specific fuel consumption s.f.c as

$$sfc = \frac{\dot{m}_f}{F_n}$$ Unit is kg/(N*s) or kg/(N*hr)

fuel to air ratio $f = \frac{\dot{m}_f}{\dot{m}_a}$

specific net thrust $SNT = \frac{F_n}{\dot{m}_a}$

$$SNT = \frac{F_n}{\dot{m}_a} = \frac{\dot{m}_f}{\dot{m}_a}\frac{F_n}{\dot{m}_f}$$

$$SNT = \frac{\dot{m}_f / \dot{m}_a}{\dot{m}_f / F_n} = \frac{f}{s.f.c}$$

Propulsive efficiency

$$\eta_P = \frac{\text{Effective power}}{\text{Increase in flow power}}$$

$$\eta_P = \frac{F_n V_a}{F_n V_a + \frac{\dot{m}_j \left(V_j - V_a\right)^2}{2}}$$

Assume $p_j = p_a$, $\dot{m}_j \simeq \dot{m}_a$

$$\eta_P = \frac{\dot{m}_a\left(V_j - V_a\right)}{\dot{m}_a\left(V_j - V_a\right) + \frac{\dot{m}_a}{2}\left(V_j - V_a\right)^2} = \frac{2V_a}{V_a + V_j}$$

We will get the same result if we use

$$\eta_P = \frac{F_n V_a}{\frac{\dot{m}_j V_j^2}{2} - \frac{\dot{m}_a V_a^2}{2}}$$

2.2.2 The by-pass engine

We assume $f << 1$ and $p_j = p_a$, where $f = \frac{\dot{m}_f}{\dot{m}_{ah}}$

$$F_n = \dot{m}_{ah}\left(V_{jh} - V_a\right) + \dot{m}_{ac}\left(V_{jc} - V_a\right)$$

$$\eta_P = \frac{F_n V_a}{F_n V_a + \frac{\dot{m}_{ah}\left(V_{jh} - V_a\right)^2}{2} + \frac{\dot{m}_{ac}\left(V_{jc} - V_a\right)^2}{2}}$$

We will get the same propulsive efficiency η_p if we use:

$$\eta_P = \frac{F_n V_a}{\frac{\dot{m}_{ah} V_{jh}^2}{2} + \frac{\dot{m}_{ac} V_{jc}^2}{2} - \frac{\dot{m}_a V_a^2}{2}}$$

2.3 Effect of altitude, Mach number and RPM of the high pressure compressor-turbine spool.

One can suggest the following scaling:

$$\frac{F_n}{p_a D^2} = f\left(\frac{V_a}{\sqrt{T_a}}, \frac{N D}{\sqrt{T_a}}\right)$$

D – engine's diameter
N – RPM of the spool
However, the variables on the right hand side are dimensional. They can be normalised as follows

$$\frac{F_n}{\delta} = f\left(M, \frac{N}{\sqrt{\theta}}\right)$$

$$\delta = \frac{p_a}{p_a(h=0)}$$

$$\theta = \frac{T_a}{T_a(h=0)}$$

$\delta = \rho/\rho_0$ amb. Airplane Propulsion Systems Airplane Aerodyn. Lan & Roskam

$\theta = T/T_0$ ρ_0, T_0 at sea level

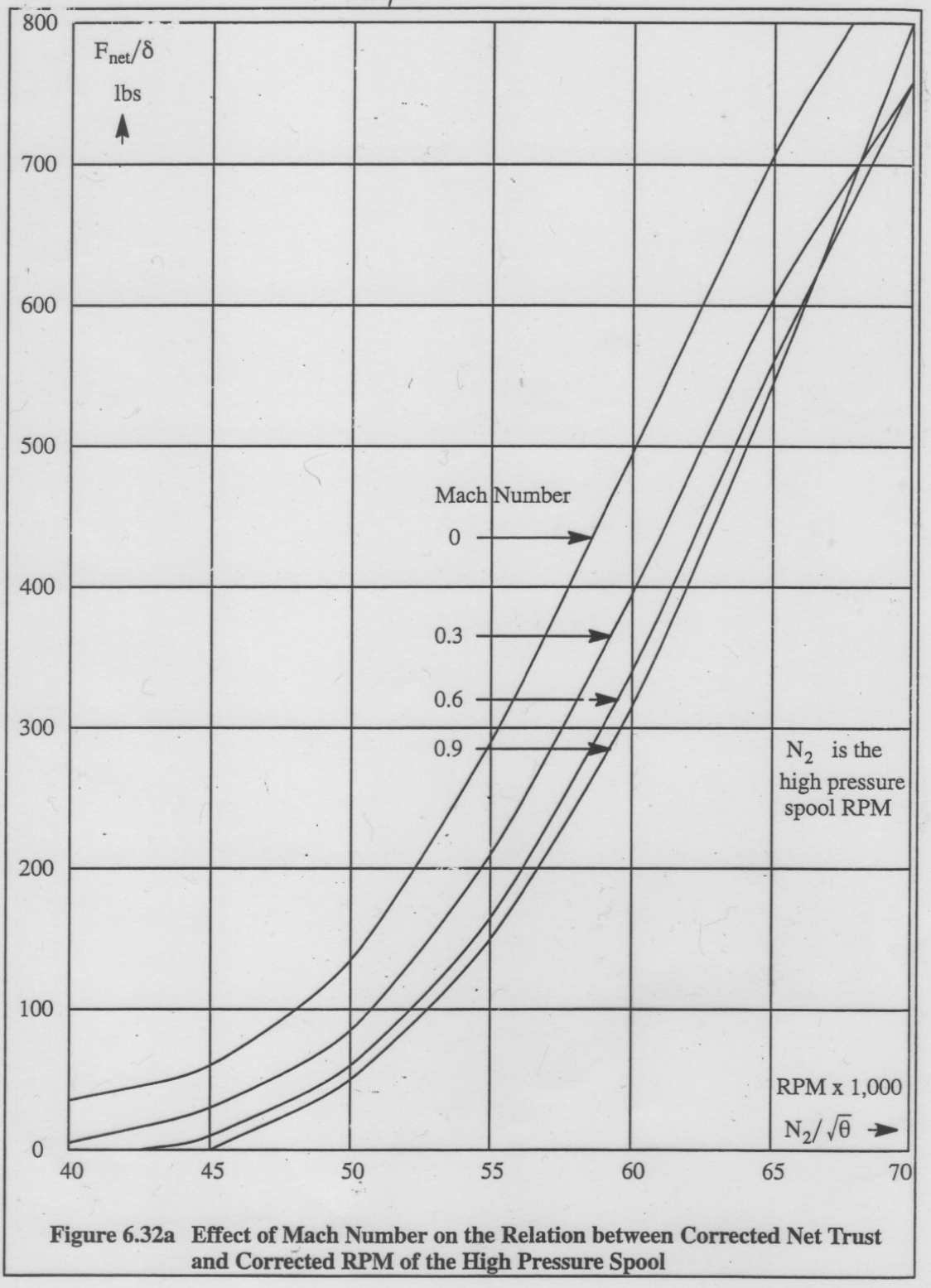


Figure 6.32a Effect of Mach Number on the Relation between Corrected Net Trust and Corrected RPM of the High Pressure Spool

2.4 Cooling turbine blades

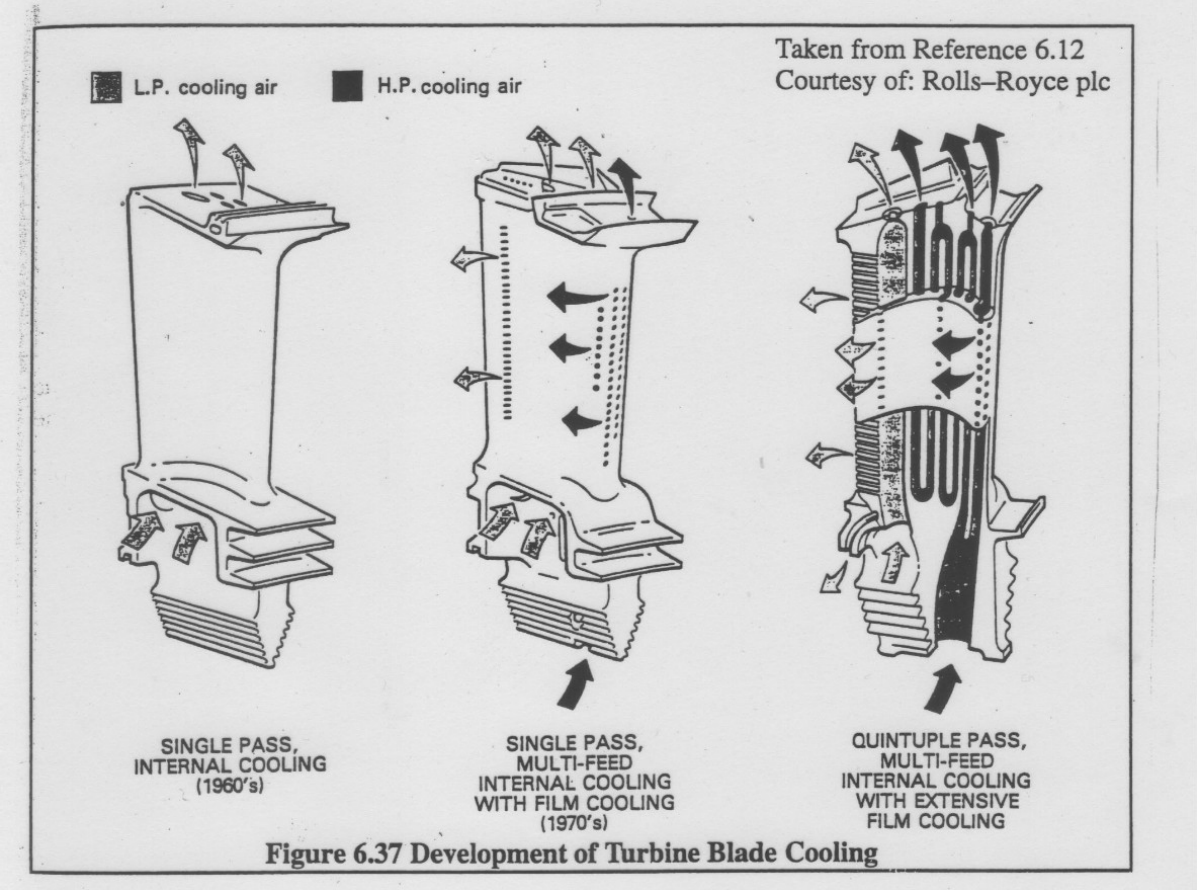


Figure 6.37 Development of Turbine Blade Cooling

High temperature in the high pressure turbine has been a major obstacle in the development of jet engines due to structure failure in the blades caused by a combination of high mechanical and thermal stresses.

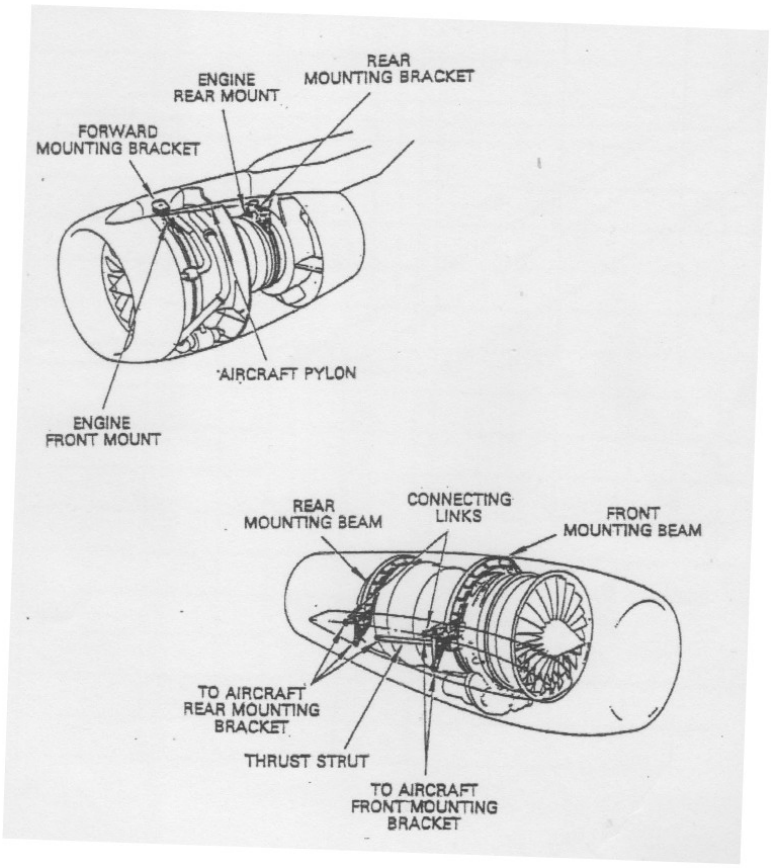
One way to mitigate this adverse effect is to bleed cold air from the compressor and direct it to cool the blades. Several methods of such cooling are illustrated above.

Other methods include redesigning the blade tip to reduce hot flow leakage between the blade tip and the casing of the engine in order to improve aerodynamic and structural performance. For further information, see:

Saleh Z, Avital EJ, Korakianitis T (2020) Effect of in-service burnout on the transonic tip leakage flows over flat tip model, *Proc Inst Mech Eng, Part A: J Power and Energy* **234**(5), 655-669

Saleh Z, Avital EJ, Korakianitis T (2021) Effect of in-service burnout on the transonic tip leakage flows over cavity tip model, *Proc Inst Mech Eng, Part A: J Power and Energy* **235**(8), 1847-1863

2.5 Engine mounting



From “Airplane aerodynamics and performance” by Lan & Roskham

The mounting has to comply to the following requirements:
1. Structural requirements
2. Easy access to the engine in terms of maintenance and quick engine replacement

Recently, interest has grown at integrating the aircraft aerodynamics with the propulsion to increase aircraft performance and thus affecting the engine mounting. An historical (unintended) example is the propeller driven aircraft that when mounted on the wing could delay stall.

2.6 Scaling laws

The engine performance depends on what is inside. However, similarities in technology between the various manufacturers allows to plot a statistical relation between the engine dry weight and take-off thrust for by-pass engines and power for turbo-props.

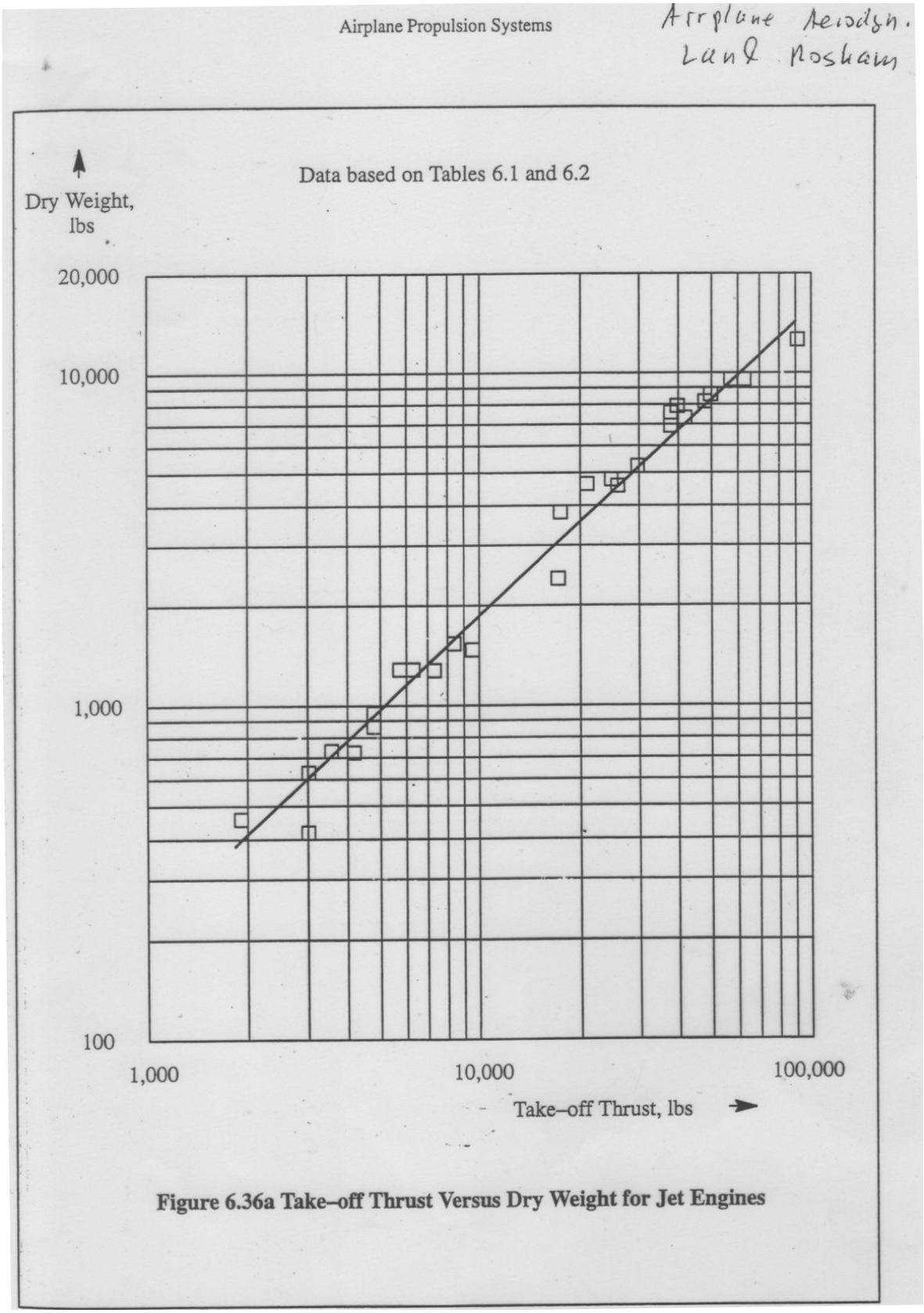


Figure 6.36a Take–off Thrust Versus Dry Weight for Jet Engines

Airplane Aerodyn.
Lan & Roskam

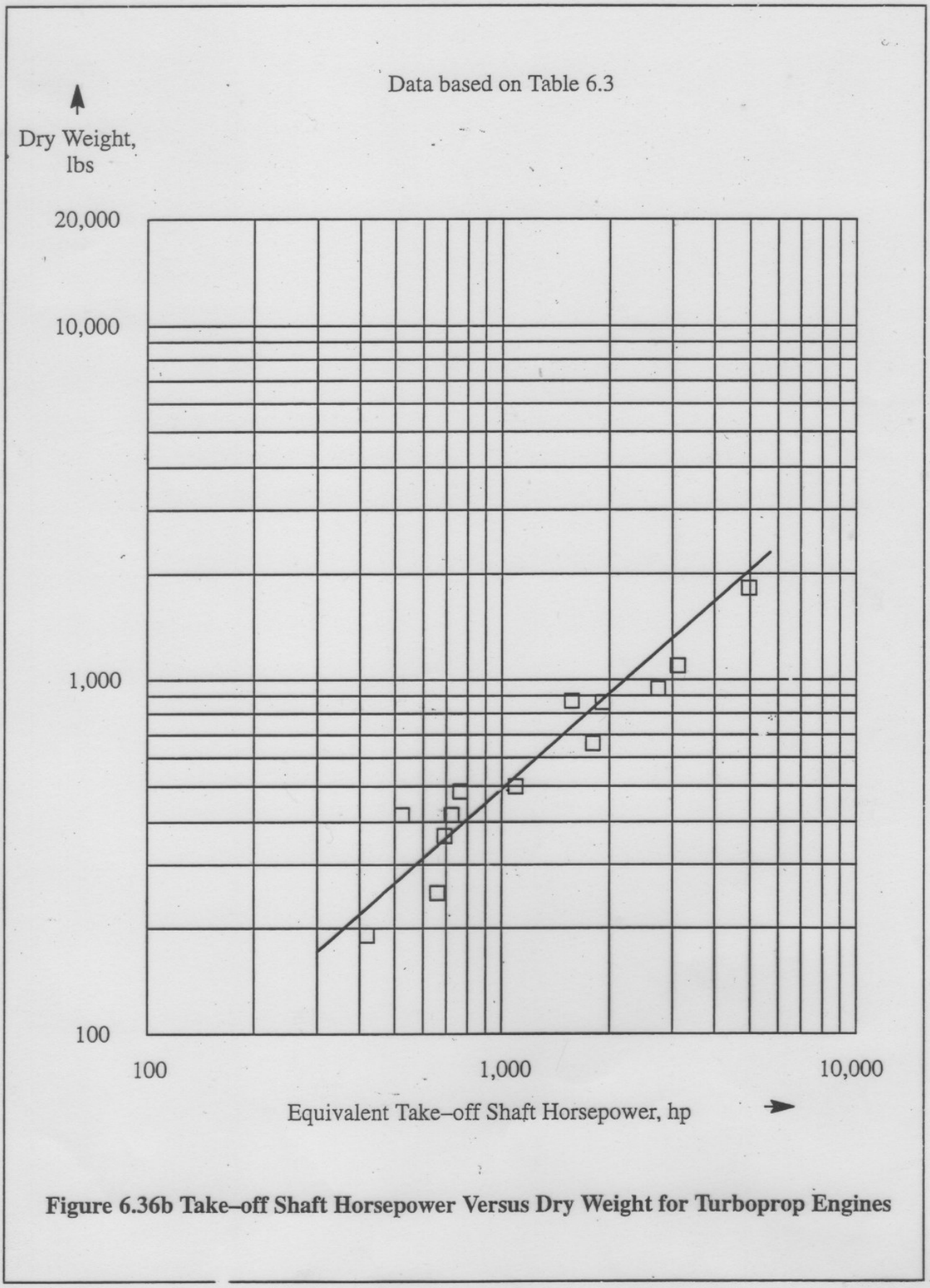


Figure 6.36b Take–off Shaft Horsepower Versus Dry Weight for Turboprop Engines

Further similarity laws can be derived as follows:

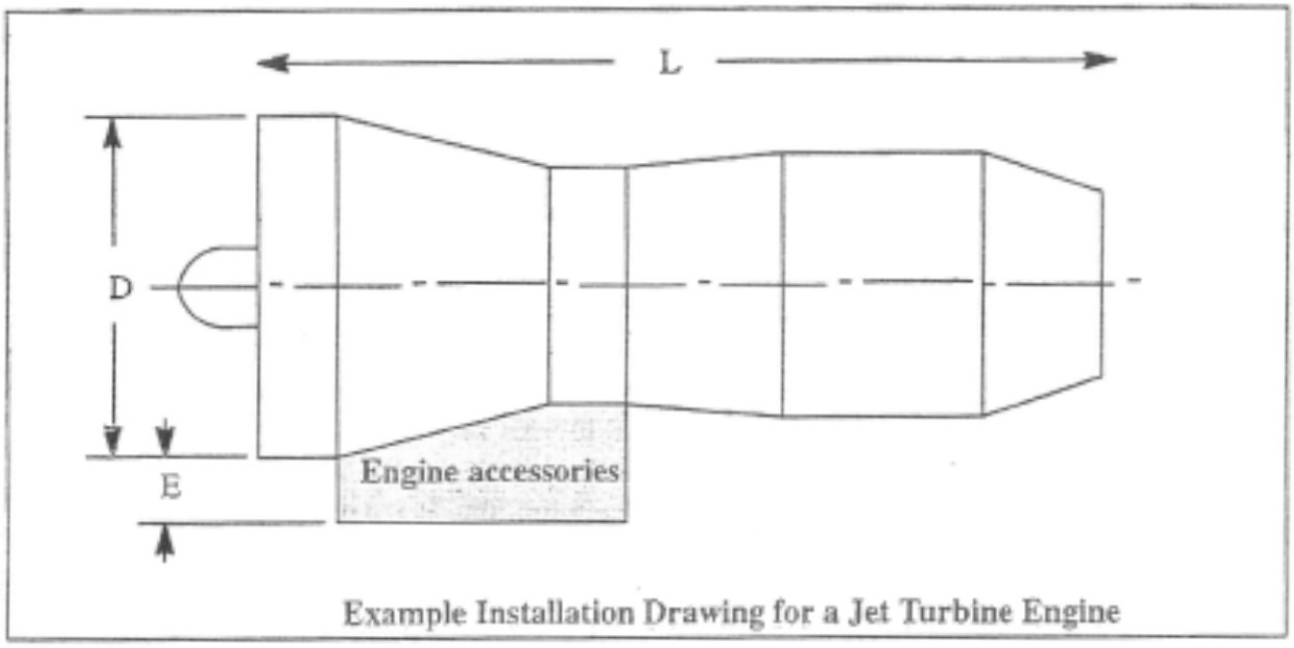


Figure 6.47 from Lan & Roskam (2008)

A) Scaling by a thrust ratio

If a new engine is required to have a certain new thrust F_{reqd} for a certain flight condition and the thrust of an actual engine F_{data} is known for the same flight condition, then the following scaling laws may be used.

$$L_{reqd} = L_{data}(SF)^{0.4} \ , \ D_{reqd} = D_{data}(SF)^{0.4} \ , \ W_{reqd} = W_{data}(SF)^{1.1}$$

Where

$SF = F_{reqd}/F_{data}$

W is the weight of the aircraft

E can be estimated as 0.25D

B) Scaling by maximum take off thrust (F_{TO}) and maximum level flight Mach number (M_{max}):

All units in the following scaling laws are imperial:
sfc – lbs/(lbs*hr), thrust – lbs, length – ft, weight – lbs

B1: No re-heat system:

$$sfc_{take-off} = 0.67\, e^{-0.12\mu} \;,\; sfc_{cruise} = 0.88\, e^{-0.05\mu}$$

$$F_{cruise} = 0.60\left(F_{TO}\right)^{0.9} e^{0.02\mu} \;,\; L = 2.22\left(F_{TO}\right)^{0.4}\left(M_{max}\right)^{0.2}$$

$$D = 0.393\left(F_{TO}\right)^{0.5} e^{0.04\mu} \;,\; W = 0.084\left(F_{TO}\right)^{1.1} e^{-0.045\mu}$$

Where μ is the by-pass ratio

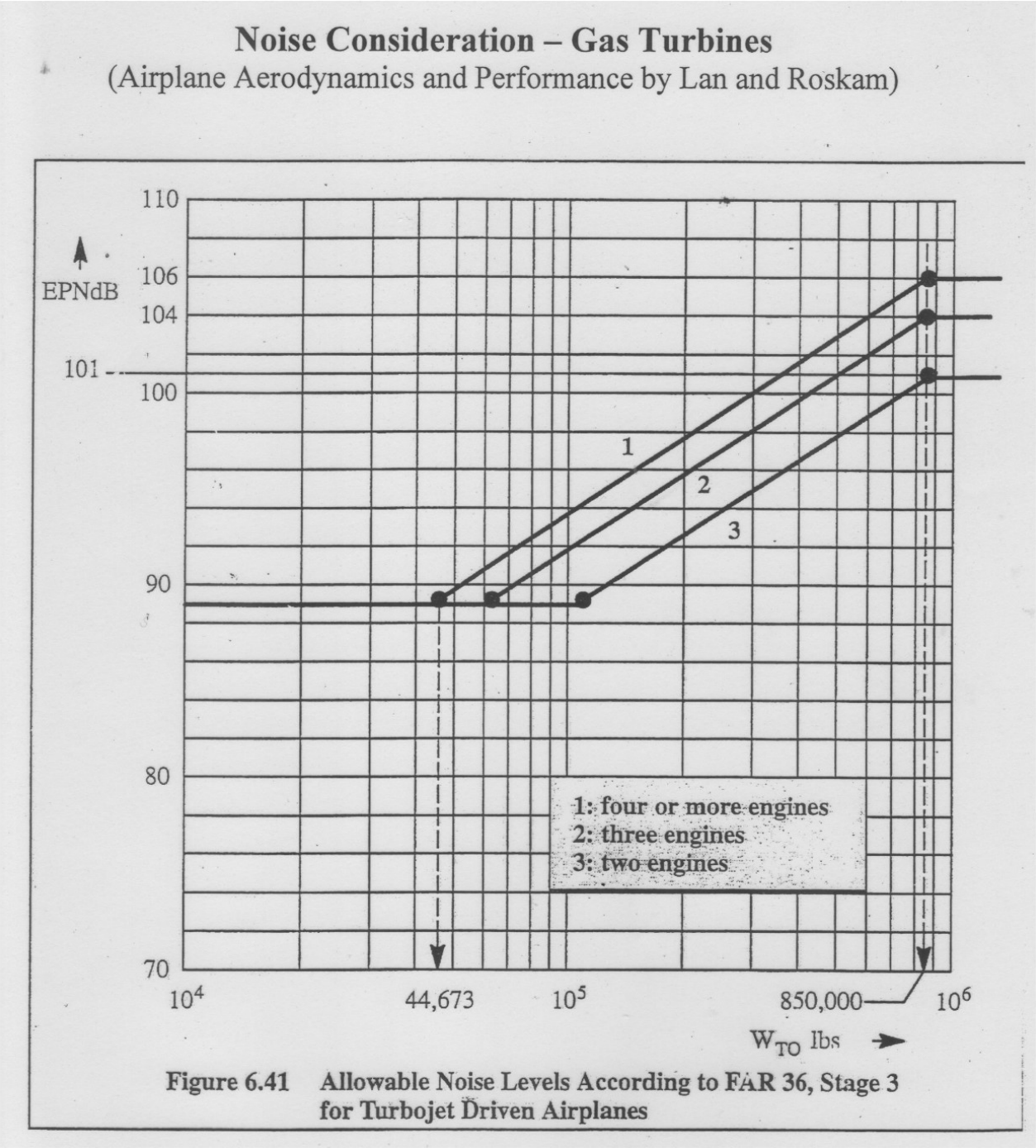
B2: With re-heat system

$$sfc_{take-off} = 2.10\, e^{-0.12\mu} \;,\; sfc_{cruise} = 1.04\, e^{-0.186\mu}$$

$$F_{cruise} = 1.60\left(F_{TO}\right)^{0.74} e^{0.023\mu} \;,\; L = 3.06\left(F_{TO}\right)^{0.4}\left(M_{max}\right)^{0.2}$$

$$D = 0.288\left(F_{TO}\right)^{0.5} e^{0.04\mu} \;,\; W = 0.063\left(F_{TO}\right)^{1.1}\left(M_{max}\right)^{0.25} e^{-0.81\mu}$$

2.8 Noise consideration



EPNdB – Effective Perceived Noise level in dB

Turbo-jet Turbo-fan

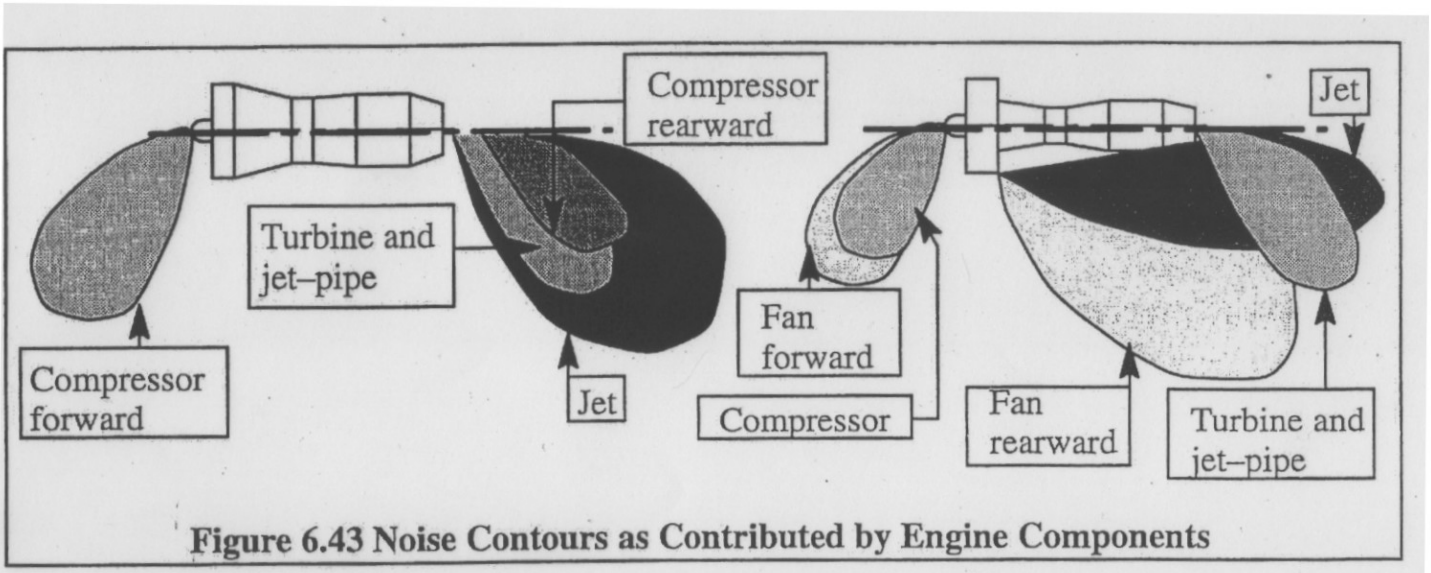


Figure 6.43 Noise Contours as Contributed by Engine Components

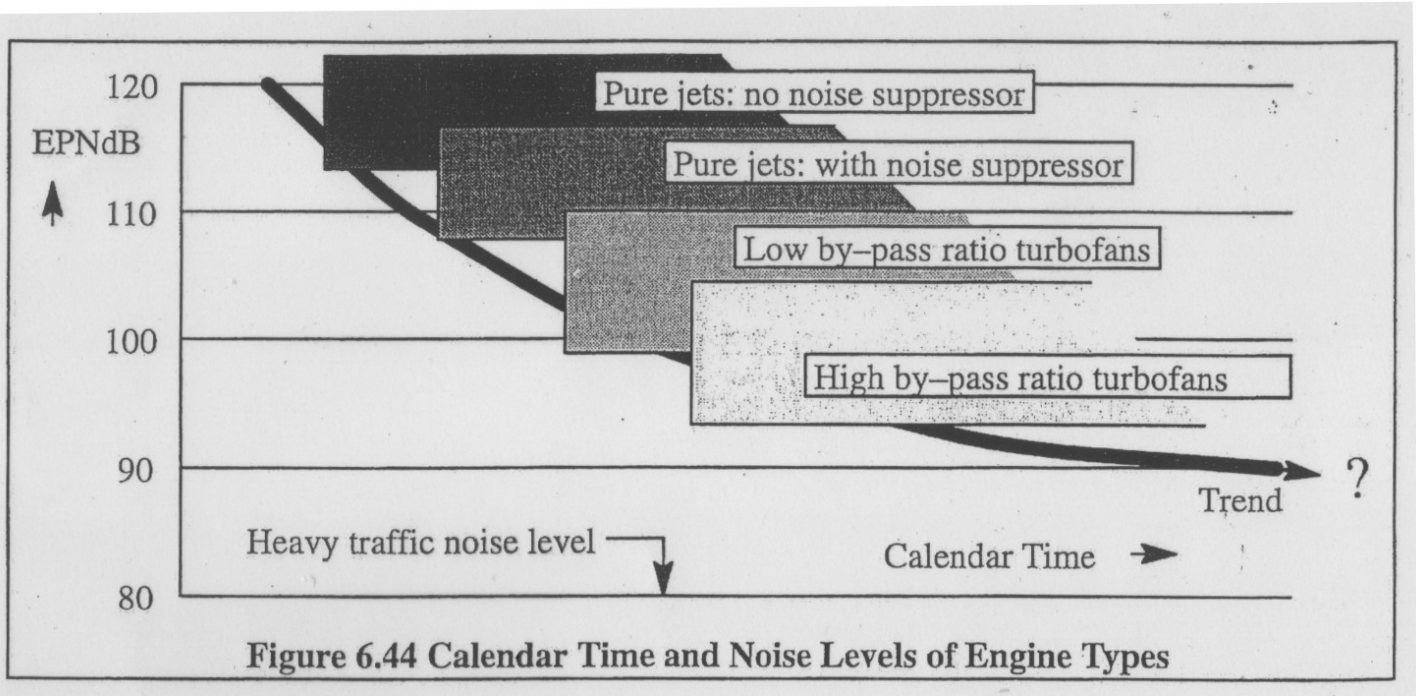
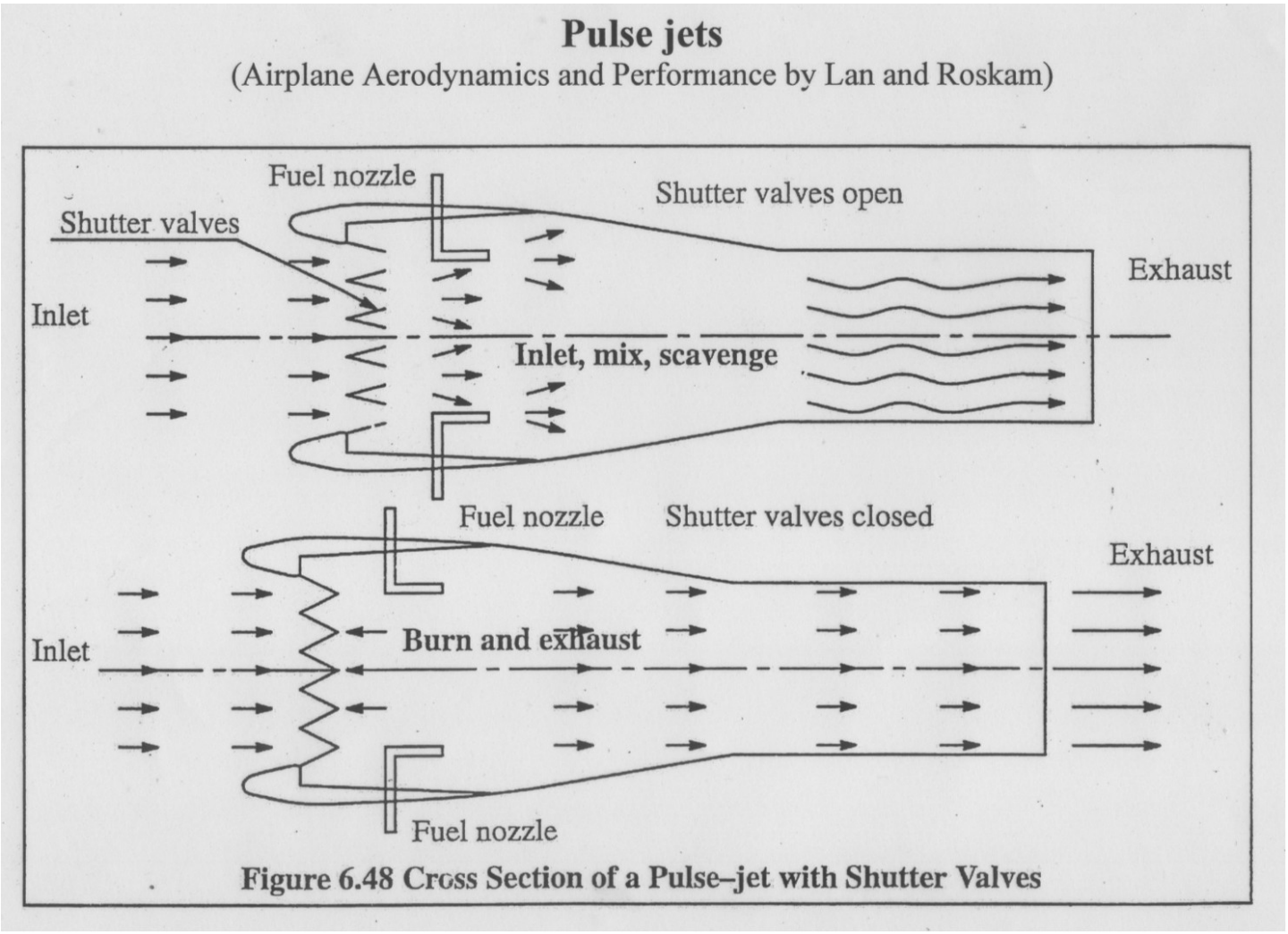


Figure 6.44 Calendar Time and Noise Levels of Engine Types

From: “Aircraft noise” by Smith

2.9 Pulse engine jet



Pulse jet engine with shutter valves:

1. Inlet valves are opened: New air enters the engine, mixes with the fuel
2. Explosion takes place: The inlet shitter valves are closed by the explosion and gas product is emitted through the nozzle

Pulse jets are relatively simple and easy to construct. However, their fuel consumption is very high as well as their noise levels. So they are not widely used. They were used for drones.